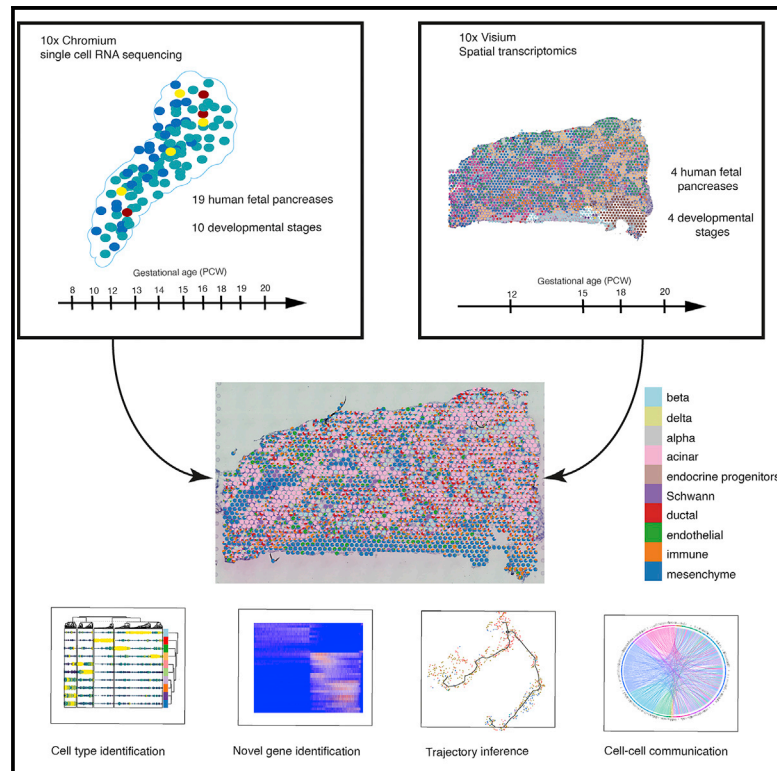


Cell Metabolism

Single-cell transcriptomic and spatial landscapes of the developing human pancreas

Graphical abstract



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In brief

By integrating single-cell and spatial transcriptomics of the human fetal pancreas across multiple developmental time points, Olaniru et al. present detailed analysis of the various cell types as they differentiate and mature from 8 to 20 weeks post-conception. The data are made freely accessible to the research community.

Highlights

- Combined scRNA-seq and spatial transcriptomics of human fetal pancreases
- Molecular spatial landscape of human pancreas development revealed
- Identification of cellular diversity and lineage dynamics through data integration
- Schwann cells co-locate with endocrine progenitors and may interact via L1CAM-EPHB2



Resource

Single-cell transcriptomic and spatial landscapes of the developing human pancreas

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SUMMARY

Current differentiation protocols have not been successful in reproducibly generating fully functional human beta cells *in vitro*, partly due to incomplete understanding of human pancreas development. Here, we present detailed transcriptomic analysis of the various cell types of the developing human pancreas, including their spatial gene patterns. We integrated single-cell RNA sequencing with spatial transcriptomics at multiple developmental time points and revealed distinct temporal-spatial gene cascades. Cell trajectory inference identified endocrine progenitor populations and branch-specific genes as the progenitors differentiate toward alpha or beta cells. Spatial differentiation trajectories indicated that Schwann cells are spatially co-located with endocrine progenitors, and cell-cell connectivity analysis predicted that they may interact via L1CAM-EPHB2 signaling. Our integrated approach enabled us to identify heterogeneity and multiple lineage dynamics within the mesenchyme, showing that it contributed to the exocrine acinar cell state. Finally, we have generated an interactive web resource for investigating human pancreas development for the research community.

INTRODUCTION

The pancreas is a multicellular organ composed of exocrine and endocrine compartments. The exocrine pancreas contains acinar and ductal cells that secrete digestive juices, while the endocrine pancreas contains alpha, beta, delta, epsilon, and pancreatic polypeptide cells that co-operatively regulate glucose homeostasis. Despite the critical roles of the pancreas in nutrient digestion and glucose homeostasis, the dysfunction of which results in pancreatitis, pancreatic cancers, and diabetes affecting more than half a billion people worldwide,^{1–3} the mechanisms underlying how the individual cell types develop in humans remain unclear.

Understanding pancreas developmental trajectories can provide essential knowledge for generating unlimited amounts of insulin (INS)-producing beta cells, for example, from stem cells for cell replacement therapies of type 1 diabetes, and much work has been expended in recent years to define effective differentiation protocols.^{4–7} However, current differentiation strategies, which are mainly based on recapitulating gene cascades identified in mouse pancreas development, do not reproducibly produce fully functional human beta cells *in vitro* that have the functional attributes of primary human islets.^{5,8–10} This is not surprising as human islet development occurs over a longer time

span than in mice and inter-species differences have been reported, such as in the islet cytoarchitecture, and the presence of a single wave of endocrine differentiation and delayed expression of key differentiation genes in humans.^{11,12} Several methods to generate mature stem cell-derived beta cells have been successfully developed such as improved beta cell differentiation and enrichment strategies;⁷ stem cell-derived islet clock entrainment,¹³ using cell-surface markers to isolate pancreatic progenitors;¹⁴ and formation of immunoprotected human-islet-like organoids.¹⁵ However, the stem cell-derived beta cells are not metabolically identical to adult human islets,¹⁶ and a clearer understanding of human pancreatic endocrinogenesis, obtained through deep profiling of developing human pancreas, may provide additional insights to optimize beta cell maturation from stem cells. Some progress has recently been made in this through single-cell RNA sequencing (scRNA-seq). Thus, different progenitor populations have been identified in the very early stages of pancreas development (7 and 10 post-conception week [PCW])^{17,18} and significant differences in lineage differentiation between developing mouse and human pancreas have recently been identified.¹⁹

However, while scRNA-seq provides a snapshot of gene expression profiles at an unprecedented scale, gene information in relation to spatial cell context is lost since tissue dissociation is



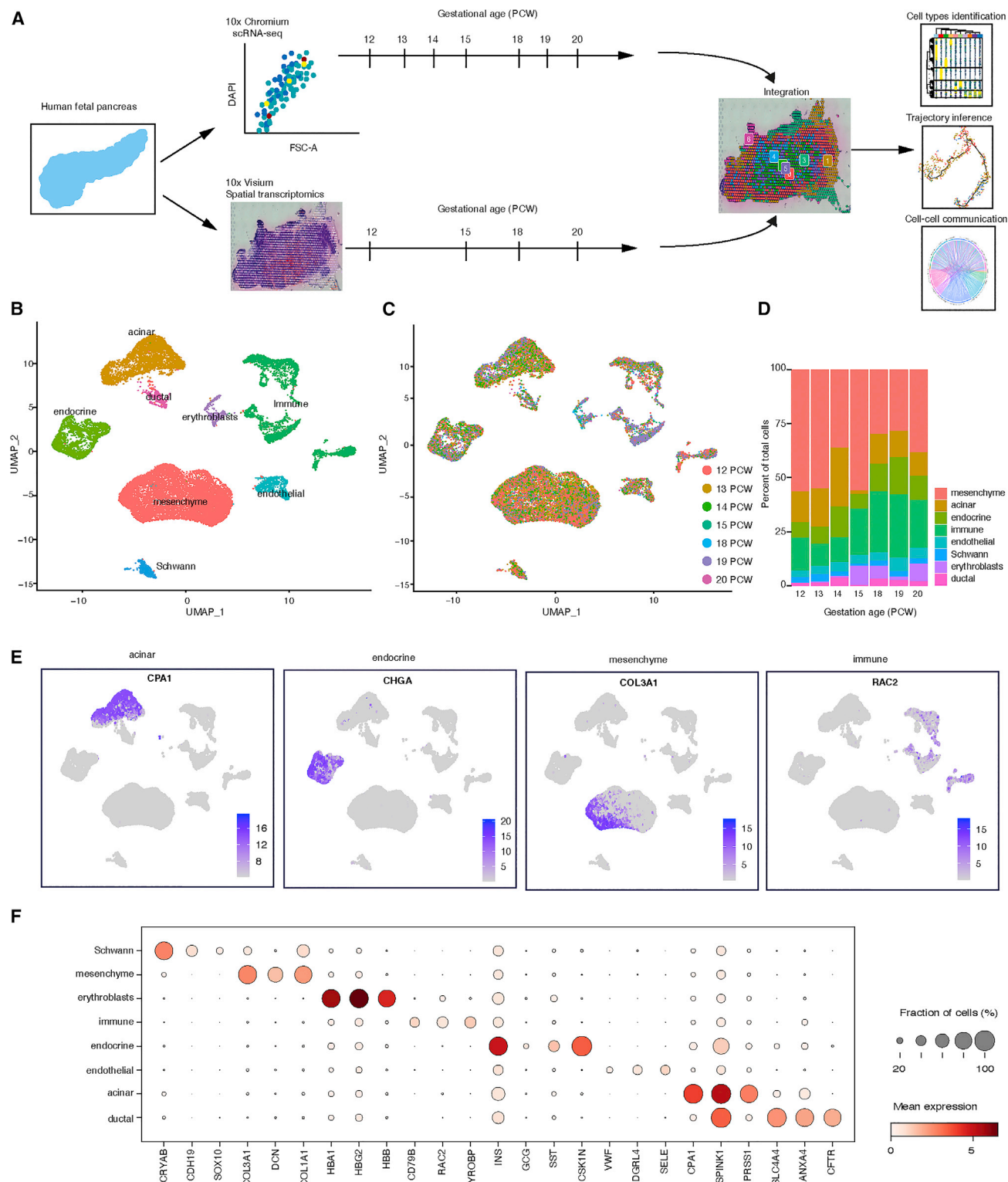


Figure 1. Single-cell sequencing of the developing human pancreas

(A) Schematic showing study design for spatiotemporal analysis of human pancreas development. Twelve whole fetal pancreases at 12, 13, 14, 15, 18, 19, and 20 PCW were profiled by scRNA-seq, and pancreases at 12, 15, 18, and 20 PCW were processed for 10x Visium spatial transcriptomics. Before integrating the scRNA-seq with spatial transcriptomics to reveal cell types and trajectories analyses, data from a previous study¹⁹ were merged with the scRNA-seq data to extend the time points from 8 to 20 PCW.

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required. Spatial transcriptomics gives positional gene patterns and provides spatial attributes of cells within the tissue context. In this study, we have therefore utilized both high-throughput scRNA-seq and spatial transcriptomics of human fetal pancreases at multiple developmental stages at 8, 10, 12, 13, 14, 15, 16, 18, 19, and 20 PCW followed by data integration to define the cellular heterogeneity and spatial developmental landscape of human pancreas development. This approach has allowed us to characterize and spatially resolve multiple human pancreatic cell populations at different developmental stages, including their cell-cell interactions, and we have identified novel gene candidates that regulate progenitor cell differentiation. By estimating pairwise similarity in transcriptional profiles among cells, we have uncovered spatial differentiation trajectories *in situ*, which enabled us to identify, for the first time, the potential importance of Schwann cells and mesenchymal cells in the differentiation of human endocrine progenitors and acinar cells, respectively.

RESULTS

scRNA-seq analysis of whole human pancreases at 12–20 PCW

We combined scRNA-seq with 10x Visium spatial transcriptomics to systematically characterize the developing landscape of the human pancreas, as indicated in Figure 1A. For scRNA-seq, whole pancreases from 12 individual embryos spanning 7 developmental time points (12 PCW, 2 embryos; 13 PCW, 3 embryos; 14 PCW, 2 embryos; 15 PCW, 1 embryo; 18 PCW, 1 embryo; 19 PCW, 2 embryos; 20 PCW, 1 embryo) were dissociated, and live-sorted single cells were multiplexed with antibody-oligonucleotide conjugates that bind to the ubiquitous surface markers β 2M and CD298,²⁰ which are expressed by the developing human pancreas (Figures S1A–S1F), in a process known as cell hashing. Cell hashing allows multiple samples to be pooled in a single scRNA-seq run, thereby reducing costs and minimizing the impact of batch effects that can arise from differences in sample handling.²⁰ We sequenced the cells using the 10x Chromium protocol and retained 24,080 high-quality cells following stringent filtering for downstream analysis (12 PCW, 4,099; 13 PCW, 7,168; 14 PCW, 3,530; 15 PCW, 1,664; 18 PCW, 2,171; 19 PCW, 3,827; 20 PCW, 1,621; Figures S1G–S1I). Unsupervised clustering of our scRNA-seq data revealed distinct populations of acinar, ductal, endocrine, endothelial, erythroblast, immune, mesenchymal, and Schwann cells (Figures 1B and 1C) in varying proportions (Figure 1D), which we identified by differential expression of established markers (Figures 1E and 1F). For example, acinar cells were identified by expression of carboxypeptidase A1 (CPA1),²¹ endocrine cells

by chromogranin A (CHGA),²² mesenchyme by COL3A1,²³ immune cells by RAC2,²⁴ ductal cells by CFTR,²⁵ endothelial cells by ADGRL4,²⁶ erythroblasts by HBB, and Schwann clusters by CRYAB expression²⁷ (Figures 1E and 1F).

Spatial map of the developing human pancreas

It can be seen from Figures 1B and 1C that our scRNA-seq analysis revealed the presence of multiple pancreatic cell types within the developing human pancreas at the early stages of the second trimester. To spatially localize them and profile the gene expression dynamics of the cells in their morphological context, we performed 10x Visium spatial transcriptomics on 8 pancreas sections retrieved at 12, 15, 18, and 20 PCW using replicate tissue sections that were approximately 100 μ m apart from each other. We sequenced the samples to a median depth of 177.5×10^6 reads (interquartile range 116.9 – 294.4×10^6), which yielded a mean of 1,692 genes per spot and 3,395 unique molecular identifiers (UMIs) per spot (Figure S2A). We integrated and jointly analyzed the replicate sections per stage, and our analysis showed that the replicate sections were highly comparable at each stage based on uniform manifold approximation and projection (UMAP) embedding and Pearson correlation (Figures S2B–S2E). Pre-processing and analysis of gene expression signatures in Seurat²⁸ revealed 3–10 cell clusters (3 clusters at 12 PCW, 10 clusters at 15 PCW, and 9 clusters each at 18 and 20 PCW), which correlated to distinct spatial locations within the tissues (Figure 2A). Annotation using marker genes indicated that some clusters contained multiple cells, as expected of the $\sim 55 \mu$ m spatial resolution available when using 10x Visium. Using this approach, we were able to stratify populations of endocrine cells (showing high expression of the islet hormones GHRL, somatostatin [SST], and glucagon [GCG]), pancreatic/endocrine progenitors (expressing NKX6-1, SOX9, and HES1),²⁹ endothelial cells (expressing VWF and ANGPT2), ductal/acinar (expressing CFTR and HES6), acinar (expressing HES6), and mesenchymal cells (expressing COL3A1 and VIM) (Figure 2B; 18 PCW samples). We then identified spatially proximal cells at 18 PCW³⁰ and found that spatial neighborhoods were shared by acinar and endocrine cells; by endocrine, ductal, acinar, and pancreatic progenitors; and also by endothelial and mesenchymal cells, suggesting that these neighboring cell populations are more likely to interact together (Figure 2C). We denoted interactions between cells of the same or different cell types as homo- or hetero-typic, respectively. Mesenchyme-mesenchyme pairs had the highest cell proximity score, indicating highly enriched cell-cell interactions, while endocrine cells and endothelial cells were predicted to preferentially undergo the highest number of homotypic interactions (Figure 2C, right). Spatial proximity profiling also indicated that the

(B) UMAP embedding showing different annotated cell clusters in the developing human pancreas at 12–20 PCW from 12 biological replicates following scRNA-seq.

(C) UMAP embedding of the developing pancreas transcriptomes showing the different gestational ages (PCW). The distribution of the color-coded gestational stages across clusters show that batch effects were absent following integration.

(D) Percentage frequency of cell populations in the scRNA-seq data between 12 and 20 PCW.

(E) Feature plots showing some selected marker genes that were used to annotate the major cell clusters: CPA1 for acinar cells, CHGA for endocrine, COL3A1 for mesenchyme, and RAC2 for immune cells.

(F) Dotplot showing marker gene distributions across the different cell populations.

See also Figures S1 and S4.

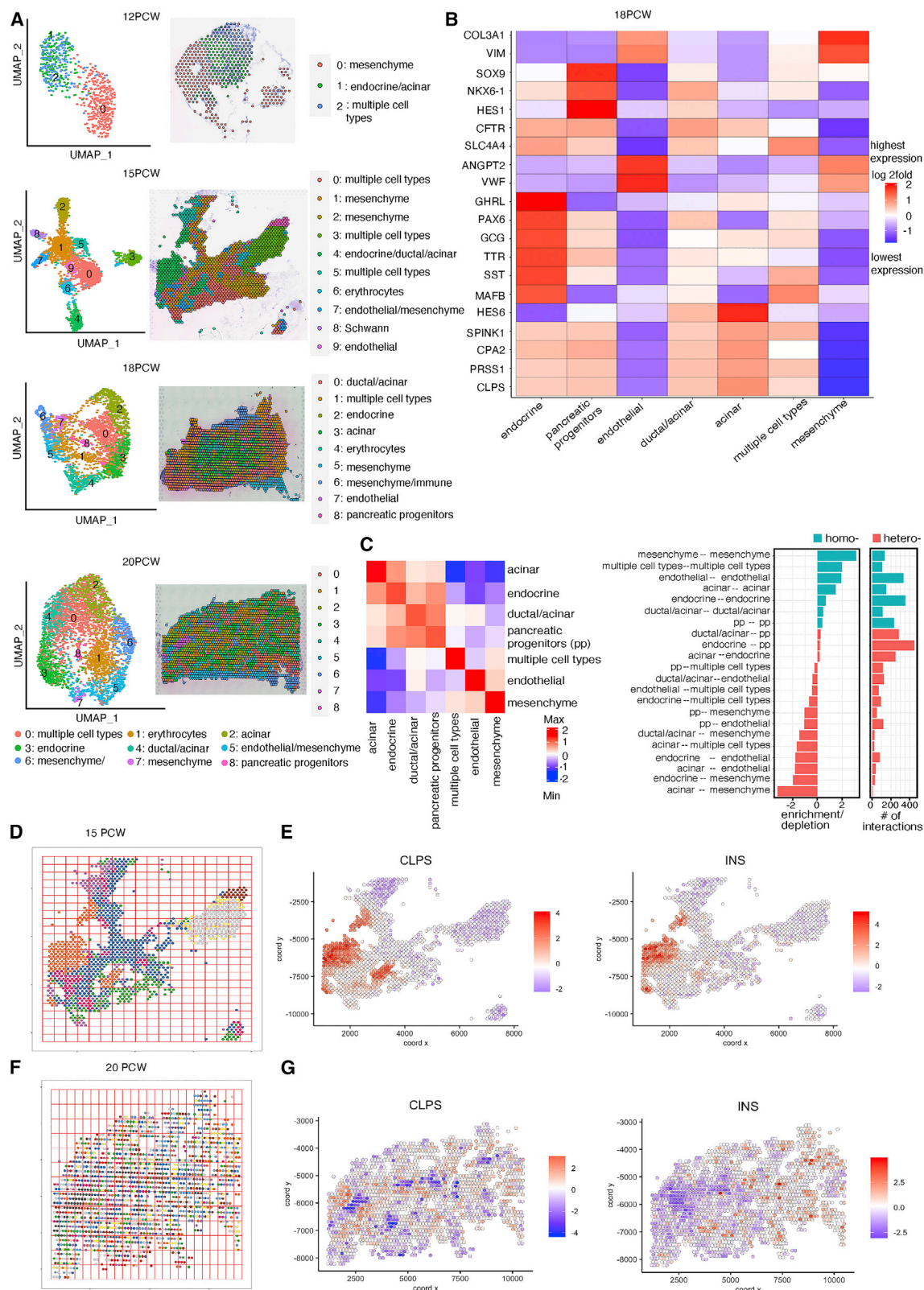


Figure 2. Spatial transcriptomics and cell-cell proximity map of the developing human pancreas

(A) UMAP embedding and spatial projection of clusters on tissue slides at 12, 15, 18, and 20 PCW. Replicate sections at each stage were integrated and jointly analyzed to produce the UMAP embedding shown.

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highest heterotypic interactions occurred between endocrine cells and pancreatic progenitors, with substantial interactions also predicted between ductal/acinar cells and pancreatic progenitors and also between acinar cells and endocrine cells (Figure 2C, right). We created spatial networks among the different cell types by connecting neighboring cells through a Delaunay triangulation implemented in Giotto,³⁰ and we defined a hub region as an area with the highest number of neighboring cells expressing the same gene. This allowed us to identify spatially variable genes driving spatial trends across the pancreas, which correlated to distinct hub regions at 15 PCW (Figures 2D and 2E) that became less prominent at 20 PCW as the pancreas expands and the cells intermingle (Figures 2F and 2G). Thus, it can be seen at 15 PCW that there was co-localization of expression of colipase (CLPS), which codes for the acinar protein CLPS, and INS (Figure 2E), indicating that at this stage of development the exocrine and endocrine pancreas are not yet defined by the distinct anatomy that is observed post-natally. However, by 20 PCW, the acinar and endocrine cells were spatially separated and discrete clusters of INS-expressing cells were evident, which most likely reflects the presence of islets scattered within the pancreas (Figure 2G). The majority of the spatially correlated genes that we identified are established canonical markers of the endocrine (INS, GCG, and proprotein convertase subtilisin/kexin type 1 inhibitor [PCSK1N]), mesenchyme (COL1A2 and COL3A1), and acinar cell populations (CEL and CPA1) (Figures S3A–S3C), but other novel spatially correlated genes were also identified. For example, ACOT7 and ATP1A1 were spatially correlated with Schwann cell populations; DCN, ADAM33, and COX6A1 with the mesenchyme; and ACTA2 with endothelial cells (Figures S3D and S3E).

Cell-type deconvolution of the spatially resolved developing human pancreas transcriptome

While the spatial transcriptomics analysis provided novel information on cell-cell proximity in the human pancreas at different stages of development (Figure 2), the 55 μ m spot diameter of 10x Visium transcriptomics does not allow single-cell resolution. We therefore combined our 10x Visium data with scRNA-seq data to characterize developing human pancreatic cell types at each spatial voxel, an approach that has recently been used to map the human endometrium³¹ and the developing chicken heart.³² We first integrated our scRNA-seq data (Figure 1) with a recently published dataset on the developing human pancreas at 8–19 PCW¹⁹ using regularized negative binomial regression²⁸ to increase the temporal resolution (Figure S4A). To distinguish the two datasets, we refer to the integrated data as “combined scRNA-seq.” Following filtering and quality control, we retained 53,204 cells from 8 to 20 PCW. We found a strong correlation be-

tween the two datasets ($r = 0.8$) (Figure S4B), with embedding values showing that the data are comparable following integration (Figure S4C), and combining the data allowed us to identify the clusters we had already found in our scRNA-seq dataset (Figure 1), but with improved resolution (Figures S4D–S4F). The combined scRNA-seq data containing all the relevant pancreatic cell types were used for deconvolution as aggregated single-cell datasets have been shown to improve deconvolution accuracy.³³ We deconvoluted the cell-type composition at each spatial spot of the 10x Visium samples by projecting the combined scRNA-seq and spatial transcriptomics data into a common latent space, and then we identified anchor cells that shared sufficient neighborhood by canonical correlation analysis.³⁴ This allowed the transfer of cell annotations in the scRNA-seq data to the spatial transcriptomics data, thereby identifying alpha, beta, delta, endocrine progenitors, acinar, ductal, endothelial, Schwann, immune, and mesenchymal cells *in situ* (Figure S5). We then incorporated the cell-type predictions into a deep-learning-based method³⁵ to visualize the cell composition within each spatial voxel and predict cell proportions at each stage of development (Figure 3). As expected, we found an expansion in epithelial cells as the pancreas develops (e.g., a 175% increase in acinar cells at 20 PCW compared with 12 PCW) and a corresponding decrease in the proportion of mesenchymal cells (Figures 3A–3E). Consistent with this, immunostaining for the epithelial marker cytokeratin 19 (KRT19) showed increased epithelial cell populations as the pancreas matured from 12 to 20 PCW (Figure S6). Mesenchymal cells were mostly located at the pancreas periphery and most of the cell types existed in hub regions at 12 PCW, which was more prominent at 15 PCW (Figures 3A and 3B). For example, distinct regions of acinar, immune, mesenchyme, endocrine/endocrine progenitors, and Schwann cells were evident at 15 PCW (Figure 3B), while by 20 PCW the cells were distributed throughout the pancreas, the bulk of which were acinar (Figure 3D). Among the endocrine cells, alpha cells were more than twice as abundant as beta cells in a 12 PCW section, but in another 12 PCW section and at 15, 18, and 20 PCW there were comparable proportions of these two cell types (Figure 3E). It is possible that there is further expansion in beta cell numbers as gestation extends beyond 20 PCW, thus resulting in higher numbers of beta cells than alpha cells post-natally. Adult human islet cell compositions are heterogeneous and donor dependent, with reports of 48%–59% beta cells and 33%–46% alpha cells,³⁶ 28%–75% beta cells and 10%–65% alpha cells,³⁷ and 25%–80% beta cells and 2%–67% alpha cells.³⁸ At all developmental stages that we interrogated, immune cells were spatially co-located with mesenchymal and endothelial cells but not with the endocrine cells (Figures 3A–3D), while Schwann cells were

(B) Marker gene distribution and manual annotation of clusters at 18 PCW.

(C) Heatmap (left) and bar chart (right) showing spatial proximity enrichment or depletion of cell-type pairs and cell-cell interactions at 18 PCW. Spatial proximity enrichment or depletion between cell types was calculated by dividing the observed cell-cell proximity interactions by expected frequency of interactions, which is the average frequency obtained from 2,000 spatial network simulations.

(D) Spatial grid containing cells based on their spatial locations at 15 PCW.

(E) CLPS and INS gene expression at 15 PCW showing exocrine and endocrine cells are not yet fully separated.

(F) Spatial grid containing cells based on their spatial locations at 20 PCW.

(G) CLPS and INS gene expression indicating that exocrine and endocrine cells are spatially separated at 20 PCW.

See also Figures S2 and S3.

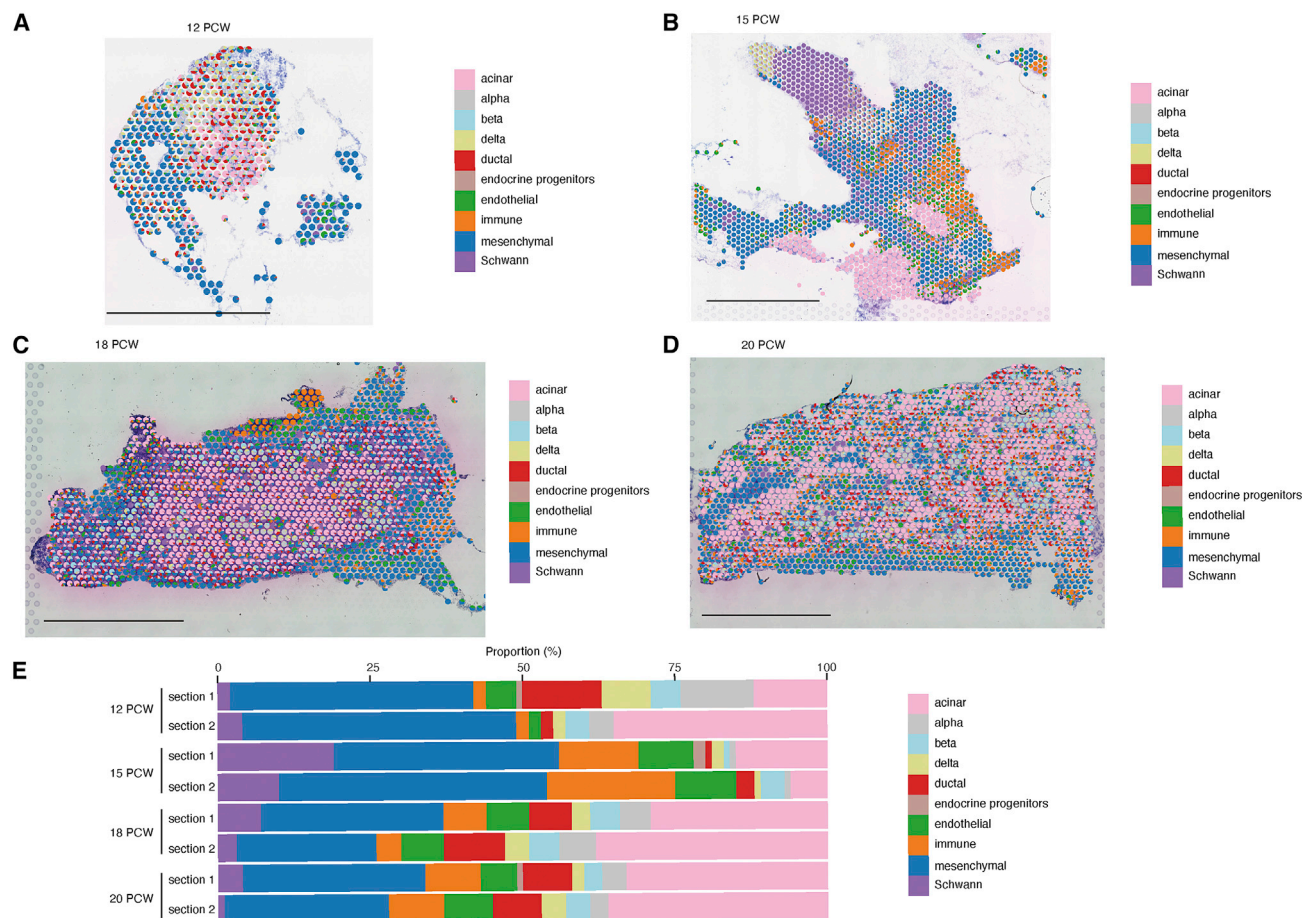


Figure 3. Cell-type map of the developing human pancreas

(A–D) Cell-type prediction overlay on ST spots showing the cells inferred from scRNA-seq at (A) 12 PCW, (B) 15 PCW, (C) 18 PCW, and (D) 20 PCW. Scale bar at all stages, 2 mm.

(E) Stacked plot showing the proportion of cells inferred for each section at the different developmental stages (12–20 PCW).

See also [Figures S4–S6](#).

in close spatial proximity to mesenchymal cells and endocrine progenitors at 15 PCW ([Figures 3A and 3B](#)).

Lineage dynamics within the endocrine compartment at 8–20 PCW

Having identified the various cell types in the scRNA-seq and ST datasets, we next focused on gene expression trajectory analysis of the endocrine cells. Reclustering the endocrine component of the combined scRNA-seq data led to identification of 20 sub-clusters ([Figure S7A](#)) spread across the 10 developmental time points ([Figure S7B](#); $n = 5,905$ cells). Using differentially expressed genes and canonical markers ([Figure S7C](#)), we identified distinct populations of endocrine progenitors (NEUROG3+), beta cells (INS+), delta cells (SST+), alpha cells (GCG+), pancreatic polypeptide cells (PPY+), and epsilon (GHRH+) cells ([Figure 4A](#), left). To understand the lineage relationships between the endocrine cell types, we initially projected the cells to the first two principal components and revealed a bifurcation from endocrine progenitors into two lineages ([Figure 4A](#), right) indicating that endocrine progenitors are precursors to all the endocrine cell types, as expected. We

identified four sub-clusters within the NEUROG3+ endocrine progenitor populations: early endocrine progenitors (early EPs) expressing high levels of NEUROG3, GPR160, and GHRH (but low ALDH1A1 expression); mid EP cells expressing SCG2, ALDH1A1, which is required for endocrine differentiation,³⁹ and the alpha cell transcription factor IRX2; late EPs with high levels of FEV, PAX4 and KCNK17 expression; and a pre-beta cell cluster expressing INS, MAFB, and ERO1B but low levels of NEUROG3 ([Figure S7D](#)). Slingshot trajectory analysis⁴⁰ corroborated our annotation by placing the early EP cluster at the beginning of the pseudotime, which corresponds to the early stages of human pancreas development at 8 and 10 PCW, with pre-beta cells at the end of the trajectory ([Figure S7E](#)). We found approximately 100 genes whose expression changed significantly ($qval < 0.05$) along the pseudotime from early EP to pre-beta cells ([Figure S7E](#)). These data point to a sequence of activation within the endocrine progenitors of $NEUROG3^{ON}/GHRH^{ON} \rightarrow ALDH1A1^{ON} \rightarrow PAX4^{ON} \rightarrow NEUROG3^{OFF}/MAFB^{ON}/INS^{ON}$. Our investigation of lineage relationships among the various endocrine cell types suggested that the beta cell clusters were not grouped together ([Figure 4A](#)), but visualizing the UMAP in

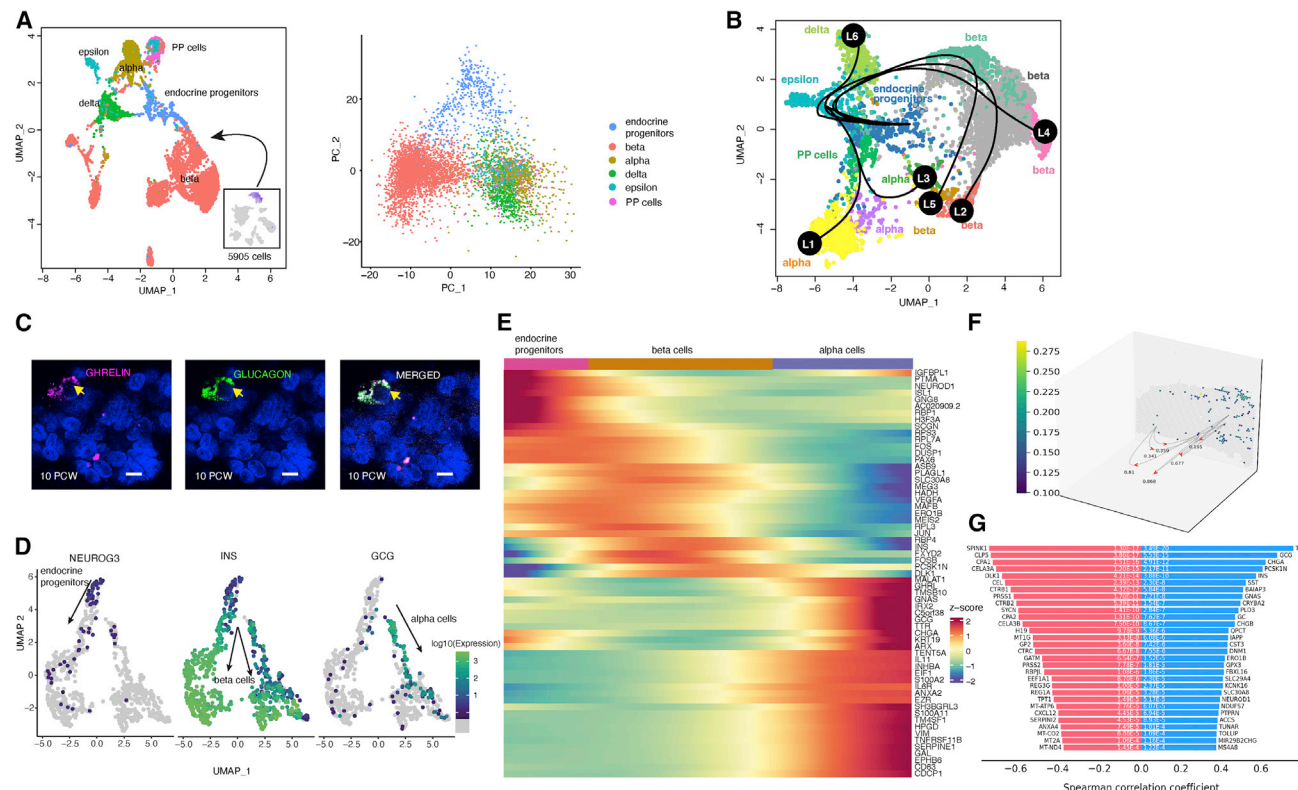


Figure 4. Identification of cell types and trajectory inference within the endocrine cluster

(A) UMAP embedding showing annotated sub-clusters of the endocrine cells. Endocrine cells from the combined scRNA-seq dataset were sub-clustered and annotated with marker genes: INS for beta cells, GCG for alpha cells, SST for delta cells, GHRL for epsilon cells, PPY for pancreatic polypeptide cells, and NEUROG3 for endocrine progenitors. Right: PCA plot showing the separation of cells within the endocrine cluster based on the first two principal components.

(B) Pseudotime analysis in Slingshot showing predicted lineages within the endocrine clusters. The lineages are predicted to arise from NEUROG3+ endocrine progenitors and are routed through GHRL-positive cells.

(C) Representative fluorescent immunohistochemical staining images of human fetal pancreas at 10 PCW showing co-localization of ghrelin and glucagon. The yellow arrowhead shows a cell expressing ghrelin and glucagon. Scale bars, 10 μ m.

(D) Pseudotime analysis in Monocle 3 showing predicted bifurcation of NEUROG3^{low}/INS^{low}/GCG^{low} endocrine progenitors into beta and alpha cell lineages.

(E) Heatmap showing branch-specific genes as endocrine progenitors differentiate toward beta and alpha cells.

(F) Pseudo-space-time analysis of an INS⁺/NEUROG3⁺ cluster showing trajectory directions.

(G) Top transition markers that are either positively correlated (blue) or negatively correlated (red) with the spatial trajectory in (F). See also Figure S7.

3D (Figure S7F) showed that this was a distortion caused by dimensionality reduction of forcing high-dimensional data into 2D.^{41,42} We therefore initially computed Slingshot trajectory analysis in 3D before, later visualizing it as a 2D UMAP. Slingshot predicted six lineages within the endocrine datasets, all arising from NEUROG3+ endocrine progenitors and suggested that GHRL+ epsilon cells may play a transitory role in these lineages (Figure 4B), consistent with the reported multipotent nature of epsilon cells in mouse pancreas development.⁴³ This was supported by immunohistochemical staining of human pancreas at 10 PCW, which indicated co-expression of ghrelin and GCG by some cells (Figure 4C). We then used Monocle 3 pseudotime analysis^{44,45} to investigate whether we could recapitulate the trajectory predictions made using principal component analysis (PCA) and Slingshot methods. Monocle 3 identified an EP population with low expression of NEUROG3, INS, and GCG (NEUROG3^{low}/INS^{low}/GCG^{low} EP) (Figure 4D), and these cells also expressed important transcription factors known to

be involved in endocrine pancreas development such as NEUROD1, INS gene enhancer protein ISL1, and PAX6 (Figure 4E). In addition, genes that have been implicated in development of other tissues were also identified, such as INS like growth factor binding protein like 1 (IGFBP1), prothymosin alpha (PTMA), and G protein subunit gamma 8 (GNG8)^{46–49} (Figure 4E). The Monocle analysis indicated that the EP cells later split into two branches, the first of which was of beta cell lineage while the second branch contained both alpha and beta cells (Figure 4D). We investigated branch-dependent gene expression as the NEUROG3^{low}/INS^{low}/GCG^{low} EP cells differentiated into alpha or beta cell lineages and identified novel branch-specific genes in addition to established markers (Figure 4E), indicating that transcriptional maturation occurred over this developmental time frame. Some of the novel beta cell genes include MEG3, ASB9, and the transcription factor MEIS2, which was previously identified as being highly expressed in beta and epsilon cells in adult human islets,⁵⁰ while predicted novel alpha cell genes

include TSMB10, TENT5A, and ANXA2 (Figure 4E). To determine whether a similar endocrine progenitor trajectory exists *in situ*, we identified cells that were positive for both NEUROG3 and INS at 20 PCW and ran spatial trajectory inference.³⁵ This allowed us to explore progression toward endocrine cells (Figure 4F), and we identified genes that were upregulated or downregulated along the spatial trajectory (Figure 4G). Known marker genes of endocrine cells such as CHGA, CHGB, transthyretin (TTR),⁵¹ GCG, SST, INS, and PCSK1N were positively correlated with the spatial trajectory while the pancreatic progenitor surface marker glycoprotein 2 (GP2)⁵² and the acinar-related genes serine protease inhibitor Kazal-type 1 (SPINK1), CLPS, CPA1, and PRSS1 were downregulated, indicating a transition of the progenitors to a more terminally differentiated endocrine state.

Tip and trunk progenitors within the human fetal pancreas

In mouse pancreas development, a subset of multipotent pancreatic progenitors residing in the central or trunk region of the pancreas epithelium are known to differentiate into ductal or endocrine cells.^{53,54} To investigate whether such progenitors with bipotent ductal-endocrine fate exist in human fetal pancreas, we sub-clustered the epithelial cells (ductal and acinar, excluding the endocrine clusters) and this revealed the presence of ductal (ANXA4+ and CTRF+), ducto-acinar (KRT19+), trunk progenitors (HNF1B+, NKX6-1+), tip progenitors (RBPJL+, PTF1A), early acinar cells (CPA1+, CEL+), mesenchymal cells (COL3A1+), and INS+ beta cells (Figure 5A). We found proliferative multipotent pancreatic progenitors at 8 and 10 PCW expressing PDX1, SOX9, HNF1B, GATA6, NKX6-1, and MKI67,^{21,29,55} with decreasing expression as gestational age increased (Figure 5B). The UMAP plot showed two visually distinct regions: one characterized by high expression of RBPJL, PTF1A, and GP2 (Figure 5C), similar to mouse tip progenitors,⁵³ and the other expressing HES1, HNF1B, and NKX6-1, similar to trunk progenitors.^{53,54} The majority (>75%) of the cells at 8 and 10 PCW are trunk progenitors, but their proportion diminished to less than 10% from 12 PCW onward, where about 25% each of the cells are tip progenitors and early acinar cells until 14 PCW (Figures 5D and 5E). The early acinar cells expanded rapidly at 16 PCW such that they represented approximately 75% of the cell population (Figure 5E). This analysis also indicated that pro-acinar genes (CPA1, PRSS1, GP2, and CTRC) are differentially expressed in tip progenitors, while pro-endocrine (MDK and SOX4) and pro-ductal genes (ANXA2) are expressed in the trunk progenitor population (Figure 5F).

The possibility of ductal cells being a source of beta cells has long been known,⁵⁶ and this has been established in mouse pancreas development,⁵⁴ *in vivo* in adult mouse pancreas,⁵⁷ and in a mouse model of INS resistance.⁵⁸ Double immunostaining of human fetal pancreas ductal epithelium and beta cells with antibodies against KRT19 and INS showed co-localization of the two proteins at 10 PCW (Figure 5G), suggesting that beta cells can also arise from ductal epithelium in human fetal pancreas. The co-localization is less pronounced at 13 PCW, but we found some cells triple positive for KRT19, CPA1, and INS (Figure 5H), suggesting that once the ductal epithelial structure is established ductal cells are less likely to differentiate to beta cells and that multipotent progenitor cells expressing CPA1, with the potential

to generate beta cells, are present in human fetal pancreas, as described in mouse.^{53,54}

Role of Schwann cells in endocrine cell development

After examining the lineage dynamics within the endocrine cluster, we next investigated the contribution of non-endocrine cells to endocrine specification. We focused on Schwann cells as they were spatially co-located with endocrine progenitor cells at 15 PCW (Figure 3B), suggesting that intercellular communications between these cells could contribute to endocrine cell differentiation. We identified eight sub-clusters in the combined scRNA-seq data, which all expressed the Schwann cell markers CRYAB and S100B (*n* = 1,761 cells; Figures S8A–S8C). We annotated cluster 0 as Schwann cell precursors (SCPs) based on expression of the stem cell marker SOX2²⁷ and cadherin 19 (CDH19),⁵⁹ while clusters 1, 4, and 7 were defined as immature Schwann cells (iSCs) as they specifically expressed GAP43⁶⁰ (Figure S8B). Clusters 5 and 6 expressed genes encoding myelin protein zero (MPZ) and proteolipid protein 1 (PLP1) (Figures S8B and S8C) and gene ontology analysis of clusters 5 and 6 identified that these cells were associated with myelination and axon development (Figure S8D), so they were designated myelinating Schwann cells (mSCs).

Given the close proximity of Schwann cells with endocrine progenitors during human pancreas development (Figure 6A), we therefore investigated cell-cell connectivity⁶¹ in our scRNA-seq data to identify ligand-receptor pairs that may participate in paracrine signaling between these populations. We identified multiple ligand-receptor pairs between Schwann and endocrine progenitors (Figure 6B) and focused on the L1 cell adhesion molecule (L1CAM)-ephrin B2 (EPHB2) interaction, as L1CAM was the most highly predicted ligand and its expression was specific to Schwann cells where it was restricted to the iSC populations (Figure S8C), while EPHB2 was expressed by endocrine progenitors and ductal cells (Figure S8E). We mapped the expression of L1CAM-EPHB2 to our spatial transcriptomics data at 15 PCW³⁵ and identified that local hot spots of L1CAM-EPHB2 interactions occurred at the interface between Schwann cells and endocrine progenitors (Figure 6C). We localized GAP43+ Schwann cells and observed close proximity with NEUROG3 endocrine progenitors using single-molecule fluorescence *in situ* hybridization (smFISH) in consecutive sections of the spatial transcriptomics tissue at 15 PCW (Figure 6D). Furthermore, L1CAM and EPHB2 mRNA expression was observed at 14 PCW, with close proximity of the two signals (Figure 6E), supporting the scRNA-seq prediction of interaction between this ligand and receptor in the developing human pancreas. These observations are consistent with Schwann cells being spatially co-located with endocrine progenitors, and further work is required to determine if this contributes to beta cell specification and maturation via the L1CAM-EPHB2 pathway.

We also used partition-based graph abstraction (PAGA),⁶² which shows unbiased lineage relationships based on gene expression among cell clusters, to investigate whether Schwann cells and endocrine progenitors shared lineage relationships. This analysis indicated that cluster 3 Schwann cells (Figures S8F and S8G) shared connected edges with cluster 9 endocrine cells, containing endocrine progenitors, beta, and delta cells (Figure S8G), and these cell populations were close to each other

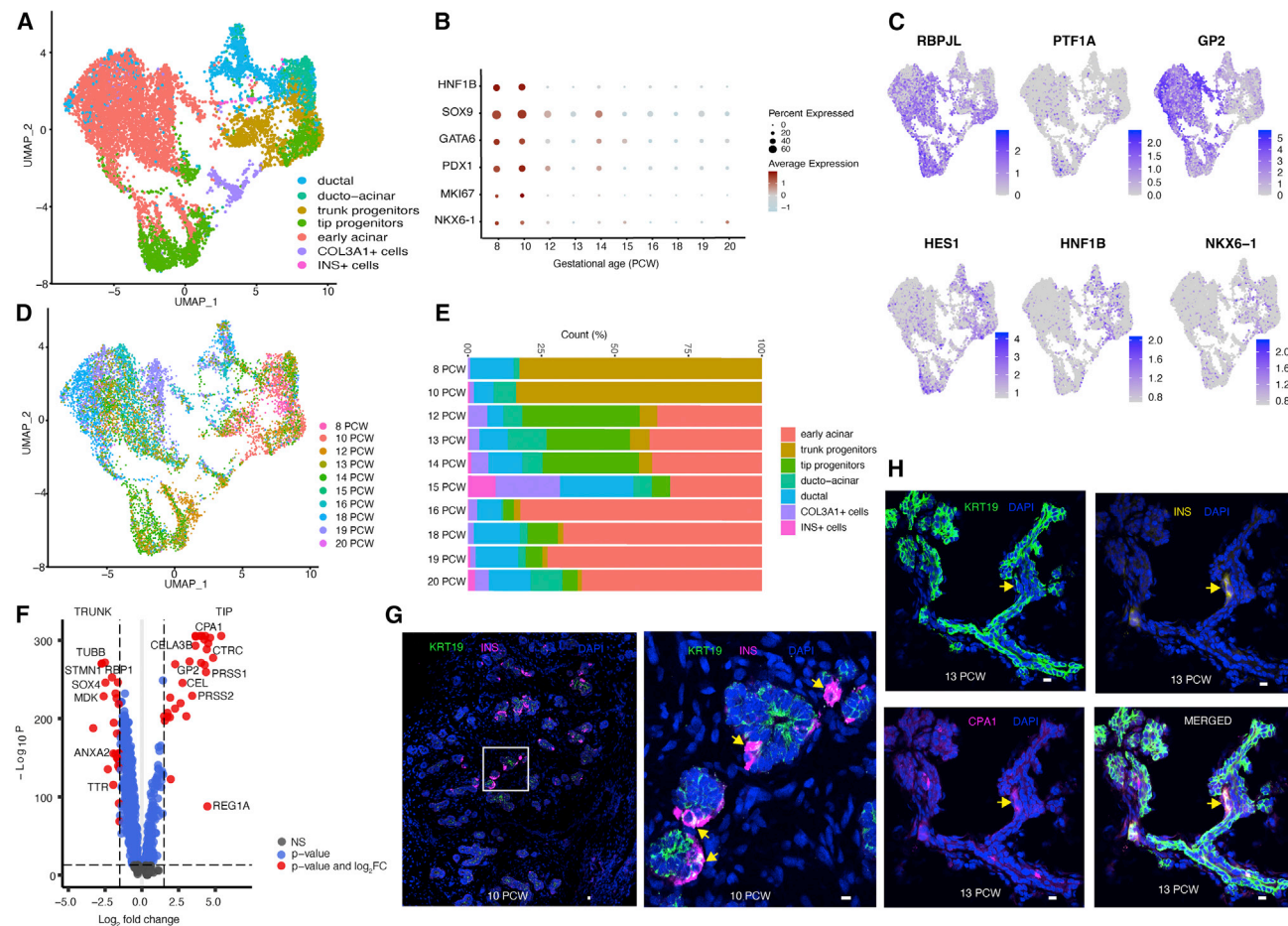


Figure 5. Identification of cell types within the ductal and acinar epithelial clusters

(A) UMAP embedding showing annotated sub-clusters of the ductal and acinar epithelial cells. Based on marker genes, clusters were annotated as ductal, ducto-acinar, trunk progenitors, tip progenitors, and early acinar. Cells expressing COL3A1 and INS were also identified.

(B) Dotplot showing expression of multipotent pancreatic progenitor genes within the epithelial cells along different gestational stages.

(C) Feature plots showing the expression of tip (upper) and trunk (lower) progenitor markers.

(D) UMAP embedding showing distribution of the different gestational ages (PCW) across the epithelial clusters.

(E) Bar chart shows the proportion of cells within indicated sub-clusters at different gestational ages.

(F) Volcano plot showing genes differentially expressed in trunk and tip progenitor populations.

(G) Representative fluorescent immunohistochemical staining images of human fetal pancreas at 10 PCW showing co-localization of cytokeratin 19 (KRT19) with insulin in some cells. Right: image represents an enlarged version of the boxed region. The yellow arrowhead shows a cell expressing cytokeratin 19 and insulin. Scale bars, 10 μ m.

(H) Representative fluorescent immunohistochemical staining images of human fetal pancreas at 13 PCW showing the expression of cytokeratin 19 (KRT19), insulin (INS), and carboxypeptidase A1 (CPA1). The yellow arrowhead shows a cell expressing KRT19, INS, and CPA1. Scale bars, 10 μ m.

within the reduced dimensionality space at 15 PCW (Figures S8H and S8I). As expected, a diffusion pseudotime plot⁶³ indicated that Schwann cells were transcriptionally earliest among all the clusters, so we set them as the root of the trajectory (Figure S8J). We then performed spatial trajectory inference³⁵ on the Schwann cell cluster and observed that endocrine cells were placed at the other end of Schwann cell trajectory (Figure 6F), with transition markers that were upregulated (blue) or downregulated (red) along the trajectory (Figure 6G). We performed gene set enrichment analysis⁶⁴ on the upregulated transition markers and found significant enrichment ($p < 0.0001$) of transcription factors involved in endocrine specification such as RFX6, PAX6, PDX1, ISL1, and NKX2-2 (Figure 6H). Transcription

factors enriched in the gene sets that were downregulated or negatively correlated with the trajectory (i.e., genes whose expression was high in Schwann cells and low in endocrine cells) included SOX2, involved in stemness;⁶⁵ POU3F1, a neuronal fate gene;⁶⁶ and the pancreatic progenitor gene HES1, suggesting that cells along the Schwann-endocrine trajectory are committed toward the endocrine lineage (Figure 6I).

Mesenchymal heterogeneity in the developing human pancreas

Despite the important structural and biochemical roles that the mesenchyme plays in mouse pancreas organogenesis,^{67,68} little is known about mesenchymal cell heterogeneity in the

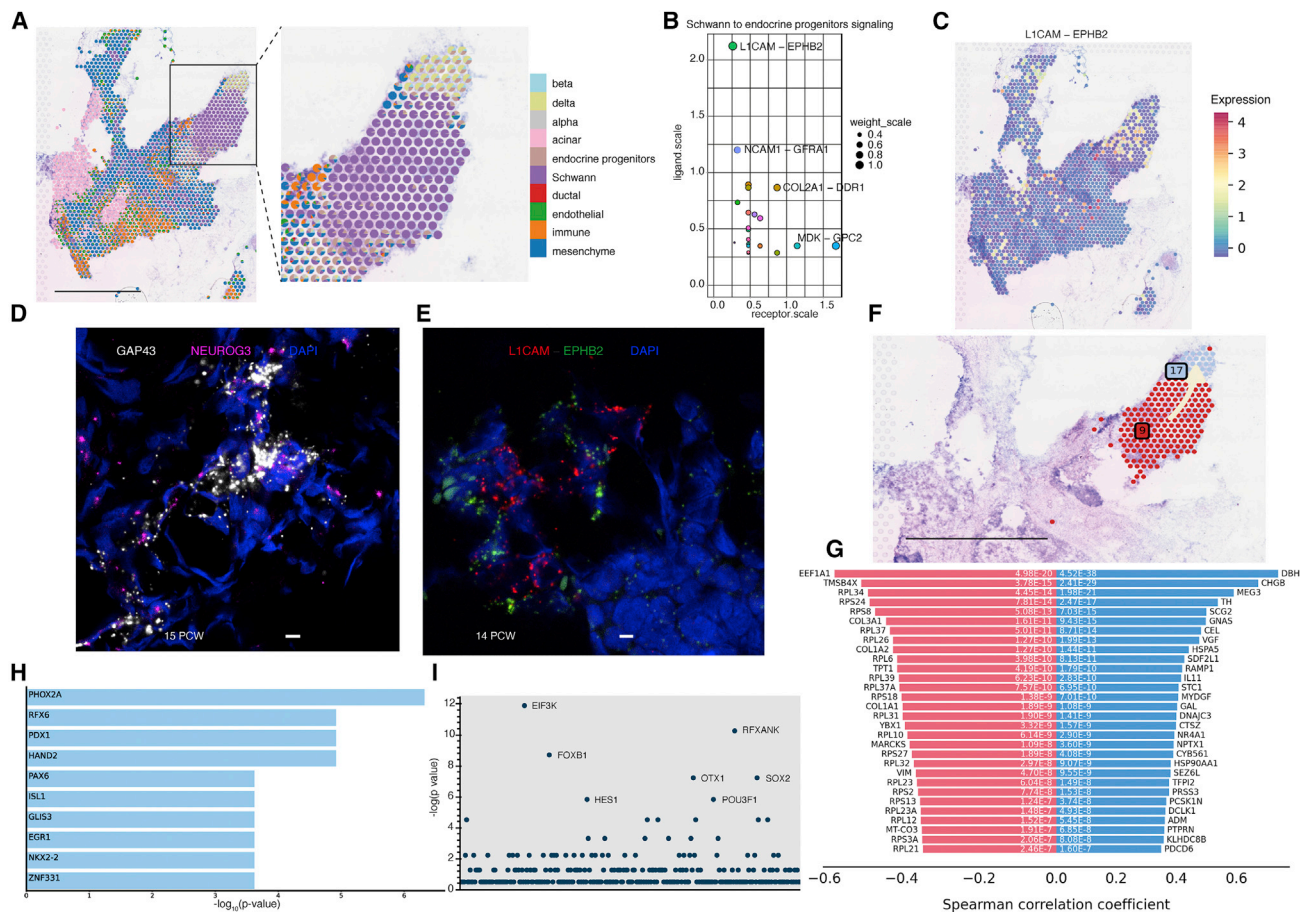


Figure 6. Role of Schwann cells in endocrine specification

(A) scRNA-seq cell-type prediction overlay on ST spots at 15 PCW. The zoomed in box shows spatial proximity of Schwann cells and endocrine cells.

(B) Ligand-receptor pairs predicted from scRNA-seq for Schwann-endocrine progenitor signaling.

(C) Spatial mapping of L1CAM-EPHB2 pairs *in situ* at 15 PCW.

(D) Fluorescence *in situ* hybridization signals for probes against GAP43 (Schwann cells) and NEUROG3 (endocrine progenitors) in 15 PCW human fetal pancreas.

(E) Fluorescence *in situ* hybridization signals for probes against L1CAM and EPHB2 in 14 PCW human fetal pancreas.

(F) Visualization of spatial trajectory between Schwann (clade 9) and endocrine clusters (clade 17).

(G) Top 30 transition markers that are positively correlated (blue) or negatively correlated (red) with the spatial trajectory between Schwann and endocrine clusters.

(H) Gene set enrichment analysis of transcription factors greater than 3-fold enriched in the positively correlated transition markers.

(I) Manhattan plot showing $-\log_{10}(p \text{ values})$ for transcription factor co-expression with the gene sets that were negatively correlated with the spatial trajectory. See also Figure S8.

developing human pancreas. We therefore re-clustered the mesenchyme in the combined scRNA-seq data (23,818 cells) and identified 19 transcriptionally distinct sub-clusters based on differential gene expression, all of which expressed varying levels of the mesenchymal genes COL1A1, COL3A1, and VIM (Figures 7A and 7B), suggesting that heterogeneity exists within them. We annotated the clusters based on expression of known marker genes and identified subtypes of myofibroblast, pericytes, vascular smooth muscle (VSM) cells, osteogenic precursors, and mesothelium (Figures 7A and 7B). We identified subtypes of stromal populations, with high expression of angiogenic chemokines⁶⁹ CXCL1, CXCL2, and CXCL3 (cluster 0); CLK1 (cluster 1); ITM2C (cluster 2); CRABP1 (cluster 4); and myofibroblasts (clusters 3 and 8) with high expression of collagens, POSTN, and DCN.⁷⁰ Cluster 17 was designated

mesothelial cells based on expression of the Wilms' tumor gene, WT1,^{71,72} KRT8, and KRT19;⁷³ clusters 15 and 16 were VSM cells since they expressed alpha smooth muscle actin ACTA2 and TAGLN;^{74,75} and clusters 7 and 12 were proliferative mesenchyme as defined by strong expression of STMN1 and MKI67, and we termed them cycling stroma. Clusters 5 and 13 were positive for the osteogenic marker CLEC11A.⁷⁶

Studies have shown that mesothelial cells can undergo transitions to form myofibroblasts, VSM, and pericytes,^{77–79} so we investigated the lineage trajectories of the mesothelial population (cluster 17). Slingshot trajectory inference identified eight lineages from the mesothelial cells, and they are predicted to form myofibroblasts (lineages L3 and L6), pericytes (L1 and L5), and VSM with myofibroblasts as intermediary cells (L2, L7, and L8) (Figure 7C). We fit a generalized additive model⁸⁰ on gene

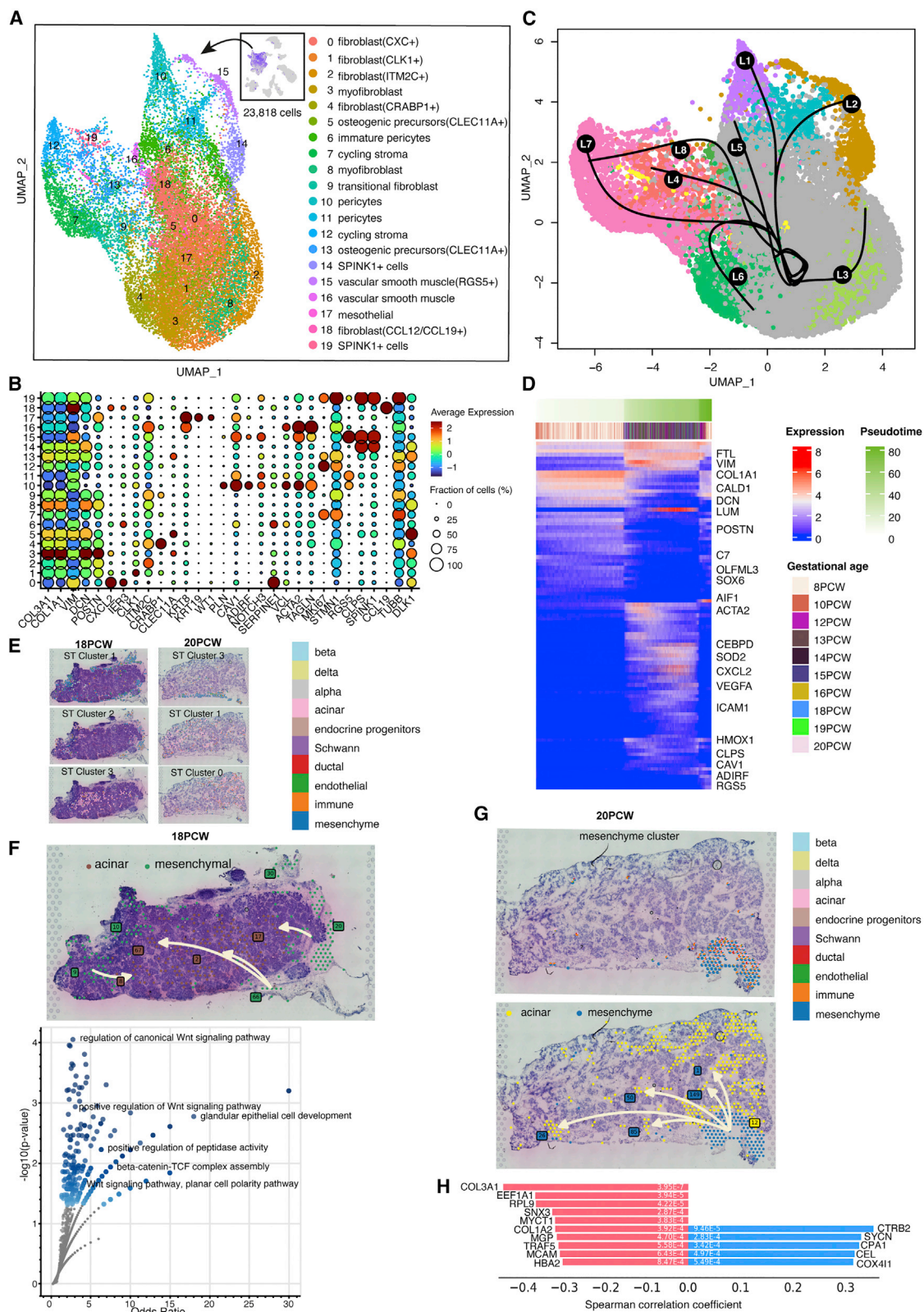


Figure 7. Heterogeneity and lineage predictions of the mesenchyme in the developing human pancreas

(A) UMAP plots showing mesenchymal cells from the combined scRNA-seq dataset that were sub-clustered and annotated with the marker genes indicated in (B). The UMAPs shown are 2D projections of 3D UMAP of the clusters.

(legend continued on next page)

expression pattern along L7 and identified genes that vary with pseudotime, including genes specific for pericytes and VSM (ACTA2, CAV1, and ADIRF) (Figure 7D). This is consistent with mesothelial cells being well-established precursors for VSM,^{78,79} and they have also been proposed to contribute to the VSM lineage in mouse pancreas.⁸¹ We identified that clusters 14, 15, and 19 were enriched for the acinar genes CLPS and SPINK1 (Figure 7B). Given the co-expression of CLPS and SPINK1 within these mesenchyme clusters, we hypothesized an association between the mesenchyme and acinar maturation. We therefore investigated lineage dynamics between mesenchymal and acinar cells *in situ* using our spatial transcriptomics data. This indicated that at 18 and 20 PCW mesenchymal cells were mostly located at the periphery of the pancreas and were often in association with endothelial or immune cells, while acinar cells were adjacent to them (Figure 7E). Spatial trajectory analysis predicted a transition from mesenchymal to acinar cells at 18 PCW (Figure 7F, upper panel) and at 20 PCW (Figure 7G). We identified transition genes that were upregulated along the spatial trajectory from mesenchymal clades 9, 20, and 66 to acinar clades 8, 17, 2, and 67, respectively (Figure 7F, upper panel). Enrichment analysis⁶⁴ of the upregulated transition genes showed significant enrichment of genes involved in Wnt/beta-catenin signaling, planar cell polarity, and glandular epithelial cell development (Figure 7F, lower panel), indicative of processes involved in acinar cell development and growth.^{82,83} Our trajectory-based differential gene expression analysis at 20 PCW also identified that mesenchymal genes including COL3A1, COL1A2, and MGP were downregulated while expression of the acinar genes CPA1 and CEL was increased (Figure 7H).

DISCUSSION

Co-ordinated interactions between different cells at critical time windows is essential for organogenesis, yet our understanding of how the diverse human pancreatic cell types interact during development is limited. Using scRNA-seq and spatial transcriptomics, we identified and localized endocrine, acinar, ductal, endothelial, immune, Schwann, and mesenchymal cell types in human fetal pancreas at 12–20 gestation weeks. To improve the temporal resolution of our scRNA-seq, we integrated it with a recently published dataset on the developing human pancreas¹⁹ to have time points from 8 to 20 PCW, and we used the combined data to deconvolute the cell types in our 10x Visium samples. Our data reveal varying proportions of pancreatic cell types, with mesenchymal cells decreasing in number over developmental time coupled to a corresponding

increase in epithelial cell populations. We uncovered heterogeneity within the endocrine, Schwann, and mesenchyme compartments with predicted lineage relationships among the sub-populations. Major strengths of our study are the availability of fetal pancreases over the relatively wide developmental range of 8–20 PCW and our combination of scRNA-seq and spatial transcriptomics over this time course. This has enabled us, for the first time, to reconstruct developmental trajectories occurring *in situ*.

It has been suggested that neural crest cells, the precursors of Schwann cells, directly differentiate into islet cells in mice,⁸⁴ but rats that lack Schwann cells progenitors do not fail to develop an endocrine pancreas.⁸⁵ Nonetheless, the plasticity of SCPs in pancreas development is not fully appreciated, and they have been implicated in cell specification in other tissues.⁸⁶ We have shown here that a Schwann cell subset, the iSCs, expresses L1CAM, and they are located in spatial proximity to endocrine progenitors, with local hotspots of L1CAM-EPHB2 interactions, which we validated by smFISH. We therefore propose that the pancreatic iSCs contribute to differentiation of endocrine progenitor cells via L1CAM-EPHB2 signaling, and this may be important in improving beta cell mass since neural crest cell/beta cell co-transplantation has been shown to increase beta cell proliferation and improve normoglycemia in diabetic mice.⁸⁷

We have also uncovered underappreciated heterogeneity within the mesenchyme, which is reminiscent of the diversity of mouse pancreas mesenchyme.⁸¹ Trajectory analysis inferred the known differentiation potential of the mesothelium into VSM and pericytes. Notably, we identified mesenchymal clusters with high expression of pro-acinar genes. A mesenchyme-acinar trajectory was predicted *in situ* from our spatial transcriptomics data, and we identified significant enrichment of gene ontologies involved in acinar cell differentiation, such as positive regulation of canonical Wnt-beta-catenin signaling, planar cell polarity and glandular epithelial cell development.^{82,83} This points to the importance of the mesenchyme in appropriate differentiation of acinar cells in the developing human pancreas, which is in agreement with a previous study where mouse exocrine pancreas failed to develop in the absence of pancreatic mesenchyme.⁸⁸

In summary, we have characterized and spatially resolved multiple human pancreatic cell populations at multiple developmental stages. We have identified sub-populations of human endocrine progenitors, novel genes that may direct their differentiation to beta or alpha cell lineages, and the influence of pancreas microenvironment on endocrine progenitor

(B) Heatmap showing differentially expressed genes in the mesenchyme 19 sub-clusters. These genes were used to annotate the clusters in (A).

(C) Pseudotime analysis in Slingshot showing predicted lineages within the mesenchyme population. The lineages are predicted to arise from the mesothelium.

(D) Heatmap showing genes that vary along the pseudotime in the predicted lineage 7 (L7) in (C).

(E) Visualization of acinar and mesenchymal cell clusters at 18 and 20 PCW.

(F) Visualization of mesenchymal-acinar trajectories with tissue localization at 18 PCW. Pseudo-space-time analysis indicated three mesenchymal sub-clusters (clades 20, 66, and 9) leading to four acinar sub-clusters (clades 17, 2, 67, and 8). The lower panel indicates a $-\log_{10}(p \text{ value})$ plot of gene ontology (GO) enrichment analysis on the positively correlated genes along the mesenchymal-acinar trajectories. GO pathways with $p < 0.05$ are colored blue.

(G) H&E image of a 20 PCW pancreas showing that mesenchymal and immune cells are spatially co-located within a cluster (upper panel). Visualization of mesenchymal-acinar trajectories with tissue localization at 20 PCW. Pseudo-space-time analysis indicated one mesenchymal cluster (clade 12) leading to multiple acinar sub-clusters (clades 1, 26, 50, 85, and 149) (lower panel).

(H) Top transition markers that are positively correlated (blue) or negatively correlated (red) with the mesenchymal-acinar spatial trajectories.

differentiation. Our data also suggest that Schwann cells contribute to beta cell specification, and further work is required to determine if the L1CAM-EPHB2 signaling axis influences the specification of endocrine progenitors toward the beta cell lineage.

We have also provided an interactive web resource (<https://www.humanpancreasdevelopment.org/>) to explore the multi-dimensional data presented in this study. Further exploration of the endocrine lineage relationships and functional roles of the branch-dependent genes along the beta cell lineage will be important in optimizing protocols to generate functional beta cells *in vitro*.

Limitations of study

The integration of scRNA-seq data with spatial transcriptomics allowed us to delineate a fine mapping of the different cell types in the developing human pancreas. However, while we were able to obtain 12 pancreases spanning 12–20 PCW, and we integrated previously published data to expand the developmental time span to 8–20 PCW, pancreases at the earliest stages of gestation (12–15 weeks) are very small, limiting the amount of analysis that can be made with them. In addition, for some time points, we only had a single pancreas, so we could not determine reproducibility of the data generated at all stages of gestation. This did limit our ability to make robust conclusions based solely on the scRNA-seq and spatial transcriptomics data, but we also carried out smFISH and fluorescent immunohistochemistry to provide orthogonal confirmation of the spatial gene data. Furthermore, this study would benefit from single-cell resolution spatial transcriptomics technology with high throughput, which is dependent on advances in the field and availability of suitable technology. It should also be highlighted that our trajectory inference predictions have not been validated by lineage tracing, which is not possible in human tissues.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cmet.2022.11.009>.

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AUTHOR CONTRIBUTIONS

Research design, O.E.O. and S.J.P.; O.E.O. conducted experiments. H.V., K.F., and P.D. prepared scRNA-seq and 10x Visium libraries. O.E.O., U.K., S.K., and S.J.P. performed data analysis. O.E.O. and S.J.P. wrote or contributed to the writing of the manuscript. All authors revised the manuscript critically for important intellectual content and gave their approval for the current version to be published. O.E.O. and S.J.P. are the guarantors of this work and, as such, had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
DAPI solution	BD Pharmingen	Cat#564907; RRID:AB_2869624
PE anti-human β 2-microglobulin Antibody	BioLegend	Cat#316305; RRID:AB_492840
PE anti-human CD298 Antibody	BioLegend	Cat#341704; RRID:AB_2274458
Human TruStain FcX	BioLegend	Cat#422301
TotalSeq-B 0254 anti-human Hashtag 4	BioLegend	Cat#394637; RRID:AB_2814350
TotalSeq-B 0256 anti-human Hashtag 6	BioLegend	Cat#394641; RRID:AB_2814352
TotalSeq-B 0258 anti-human Hashtag 8	BioLegend	Cat#394645; RRID:AB_2814354
RNAscope Probe - Hs-NEUROG3-C4 - Homo sapiens neurogenin 3 (NEUROG3) mRNA	Bio-Techne Ltd	Cat#505791-C4
Opal fluorophores	Akoya Biosciences	Opal 520, PN FP1487001KT; Opal 570, PN FP1488001KT; Opal 690, PN FP1497001KT
RNAscope Multiplex Fluorescent Reagent Kit v2	Akoya Biosciences	Cat#323100
RNAscope Probe - Hs-EPHB2 - Homo sapiens EPH receptor B2 (EPHB2)	Bio-Techne Ltd	Cat#406251
RNAscope Probe - Hs-L1CAM-C2 - Homo sapiens L1 cell adhesion molecule (L1CAM)	Bio-Techne	Cat#567451-C2
RNAscope Probe - Hs-GAP43-C3 - Homo sapiens growth associated protein 43 (GAP43)	Bio-Techne	Cat#856701-C3
Rabbit monoclonal [EPR20502] to Ghrelin	Abcam	Cat#ab209790; RRID:AB_2811161
Mouse monoclonal [A53-B/A2] to Cytokeratin 19	Abcam	Cat#ab7754; RRID:AB_306048
Rabbit monoclonal [EPR12086] to Carboxypeptidase A	Abcam	Cat#ab173283
Guinea pig polyclonal anti-insulin	Abcam	Cat#ab195956; RRID:AB_2877638
Mouse monoclonal [K79bB10] to Glucagon	Abcam	Cat#ab10988; RRID:AB_297642
Rabbit polyclonal to Neurogenin3	Abcam	Cat#ab216885
Biological samples		
Human fetal pancreas	Human Developmental Biology Resource (HDBR)	HDBR projects 200456 and 200554
Chemicals, peptides, and recombinant proteins		
Trypsin-EDTA	Sigma	Cat#T3924
OCT medium	VWR Chemicals	Cat#00411243
L-15 (Lebowitz) media	Sigma	Cat#L1518
Isopentane (2-Methylbutane)	Sigma	Cat#270342
Methanol	Sigma	Cat#34860
Eosin Y solution	Clin-Tech Limited	Cat#625025
Hematoxylin Solution, Mayer's	Sigma	Cat#MHS16-500ML
Isopropanol	Merck	Cat#I9516-1L
Dulbesco's Phosphate Buffered Saline	Sigma	Cat#D8537
RPML-1640	Merck	Cat#8758-500ML
Bovine Serum Albumin	Sigma	Cat#A7906
Collagenase from <i>Clostridium histolyticum</i>	Sigma	Cat#C2674
L-glutamine solution	Sigma	Cat#G7513

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Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
Critical commercial assays		
10x Chromium Single Cell 30 GEM, Library & Gel Bead Kit v3.1	10x Genomics	Cat#1000269
Visium Spatial Gene Expression Slide & Reagent Kit	10x Genomics	Cat#1000184
Agilent RNA 6000 Pico Reagents	Agilent Technologies	Cat# 5067-1513
Visium Spatial tissue optimization kit	10x Genomics	Cat#1000191
Visium accessory kit	10x Genomics	Cat#1000194
Deposited data		
Dataset generated from a different study	https://ngdc.cncb.ac.cn/omix	OMIX: OMIX236
Single cell RNA-Seq data generated in this study	https://www.ncbi.nlm.nih.gov/geo/	GEO: GSE197064
Spatial transcriptomics data generated in this study	https://www.ncbi.nlm.nih.gov/geo/	GEO: GSE197317
Software and algorithms		
Cellranger version 6.0.0	10x Genomics	https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest
Spaceranger version 3.1.0	10x Genomics	https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest
Giotto version 1.0.3	Github, Dries et al. ³⁰	https://rubd.github.io/Giotto_site/
Seurat version 4.0.6	R CRAN	https://satijalab.org/seurat/
Monocle version 3	R Bioconductor	https://cole-trapnell-lab.github.io/monocle3/
stLearn	Github; Pham et al. ³⁵	https://github.com/BiomedicalMachineLearning/stLearn
slingshot 2.5.2	Street et al. ⁴⁰	https://github.com/kstreet13/slinsshot

RESOURCE AVAILABILITY

Lead contact

Further information and request for reagents should be directed to and will be fulfilled by the lead contact, Professor Shanta Persaud (shanta.persaud@kcl.ac.uk).

Materials availability

We did not generate any unique reagents from this study.

Data and code availability

- The raw and processed data used for this study have been deposited on GEO with accession numbers GSE197064 (scRNA-seq) and GSE197317 (spatial transcriptomics).
- Single-cell RNA sequencing and spatial transcriptomics data were analyzed using publicly available software. R and python scripts used to analyze the data can be found at <https://github.com/olaniru/human-fetal-pancreas>.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Human samples

Human embryonic and fetal pancreases were obtained from the MRC-Wellcome Trust Human Development Biology Resource, with appropriate ethical approval from London-Fulham and Newcastle & North Tyneside 1 Research Ethics Committees. Samples received from HDBR do not include information about embryos' gender nor anything that can identify the donors. Samples were transferred immediately after provision in cold Lebowitz medium (L-15) to King's College London and processed immediately.

METHOD DETAILS

Human fetal pancreas scRNA-seq dissociation protocol

On the day of retrieval, pancreases were rinsed in ice-cold PBS, any extra-pancreatic tissues were dissected out then pancreases were inflated with 1mg/ml collagenase from *Clostridium histolyticum* (C2674). The tissues were digested at 37°C for 5 min (12–14 PCW) or 10 min (15–20 PCW), gently disrupted by pipetting with a wide-bore Rainin pipette and incubated in trypsin/EDTA for an additional 5 min at 37°C. Enzyme action was stopped with 10% FBS in RPMI medium and the samples were filtered through a 30µm mesh before being centrifuged at 1100rpm for 5min. Cell pellets were resuspended in RPMI containing 10% FBS. Cell number and viability were determined using a Countess II automated cell counter. At least 400,000 cells per pancreas with viability of 94±2% were carried forward for hash-tag staining. We initially investigated whether hash-tag (HTO) conjugation is feasible in the developing human pancreas by staining for the ubiquitous surface markers β2M and CD298 to which the oligonucleotide-tagged antibodies bind.²⁰ ~450,000–2,000,000 cells were incubated for 10min in human TruStain FcX (10%v/v) to block non-specific binding before being exposed to varying concentrations (0.25, 0.5, 1.0 and 2µg/µl) of PE β2M/PE CD298 (BioLegend, mixed 1:1) for 30 min in the dark at 4°C, then washed with cold PBS containing 0.04% BSA. 0.1ng/µl DAPI was added and FACs sorting on a BD FACSaria 3 was used to determine the percentage of live β2M/CD298-positive cells. Subsequently, HTO staining was carried out with 0.5µg of β2M and CD298 oligonucleotide-conjugated antibodies (Total-seq B Hastags 4, 6 and 8; BioLegend) with ~500,000–750,000 isolated single cells. For each scRNA-seq run, about 60,000 live cells were sorted into RPMI containing 5% FBS (4°C) before proceeding to 10x Chromium library preparation and sequencing.

10x Chromium scRNA-seq library preparation and raw sequence data processing

We used the 10x Genomics single cell RNA sequencing 3' v3 chemistry for generating our scRNA-seq data. Isolated single cells at multiple gestational stages were stained individually with Totalseq B HTOs, pooled together into 4 batches (12, 13 and 13 PCW; 12, 14 and 19 PCW; 13, 14 and 19 PCW; 15, 18 and 20 PCW) and 20,000 cells from each batch were loaded on the 10x chip. Single cell mRNA and antibody (HTO) libraries were prepared according to the manufacturer's protocol and paired-end sequencing was carried out on an Illumina NextSeq 2000, except for batch 15, 18 and 20 PCW which was sequenced on a HiSeq 2500. Generation of FASTQ files and alignment of raw sequencing reads with GRCh38 human genome reference data were carried out using Cell Ranger (version 6.0.0, 10x Genomics).

Hashed sample de-multiplexing

We first filtered the HTO antibody UMI count matrices to retain only those containing 10x cellular barcodes. We then used the HTODemux function in Seurat to separate the pooled samples into their individual components, as previously described.²⁰ We obtained a subset of the UMI count matrix and HTO count matrix using cellular barcodes contained in both matrices, which were log normalised and clustered using the clara k-mediod function. A negative binomial distribution was then fit to each cluster, with a default positive threshold of 99th percentile. This threshold was used to determine whether the cells were positive or negative for the hashtag: cells positive for more than one hashtags were classed as doublets and were removed from analysis. We also performed genetic demultiplexing⁸⁹ and observed nearly 100% concordance with the de-hashing algorithm in Seurat.

10x scRNA-seq data analysis

Output from Cellranger for the four scRNA-seq batches were loaded into Seurat (version 4.0.2) and merged. After filtering out low quality reads using **nUMI > 500, nGene > 250, log10GenesPerUMI > 0.75 and mitoRatio < 0.25 thresholds**, the Seurat objects were normalised individually by regularised negative binomial regression (SCTransform) to correct for batch effect and other technical variabilities.²⁸ The data were integrated using the IntegrateData function after selecting the most variable features which were used to extract integrating anchors using FindIntegrationAnchors. We then performed principal component analysis and used the first 20 dimensions to compute a Uniform Manifold Approximation and Projection (UMAP).⁹⁰ By FindNeighbors and FindClusters commands in Seurat, cells were clustered in a graph-based approach by the Louvain algorithm. Wilcoxon rank sum tests were performed to identify differentially expressed genes in each cluster using the FindAllMarkers function in Seurat. The clusters were annotated based on marker gene expression as endocrine (INS, GCG, SST, PAX6), acinar (CPA1, PRSS1, CLPS), ductal (SLC4A4, CFTR, ANXA4), mesenchymal (COL3A1, DCN, VIM), immune (RAC2, LYZ, TRAC), endothelial (VWF, ADGRL4, ANGPT2), Schwann (CRYAB, CDH19, SOX10) and erythroblasts (HBB, HBG2, HBA1).

Integration of scRNA-seq data with previously published 10x Chromium data

The scRNA-seq generated in this study were integrated with 10x Chromium data from a previous study on human fetal pancreas development,¹⁹ which was accessed from <https://ngdc.cncb.ac.cn/omix> with accession number OMIX236. To allow direct comparison with our data we only extracted data for the whole pancreas and excluded datasets that had been enriched for epithelial cells. We preprocessed each dataset by SCTransform and used the most variable features to identify integration anchors, which were then passed to the IntegrateData function to return a Seurat object containing an integrated expression matrix for all cells. The data were jointly analyzed and visualized by UMAP. The clusters were annotated using marker genes as described above (10X scRNA-seq data analysis).

Sub-clustering of clusters of interest

Endocrine, endocrine progenitors, mesenchymal and Schwann clusters were first individually isolated using the Subset function in Seurat and re-analyzed as described above (10X scRNA-seq data analysis). Batch effect correction was carried out with Harmony.⁹¹ Endocrine sub-clusters were annotated with the marker genes *INS*, *GCG*, *SST*, *PYY*, *GHRL* and *NEUROG3*, mesenchymal sub-clusters were annotated with *COL3A1*, *COL1A1*, *WT1*, *IGFBP2*, *ACTA2*, *TAGLN*, *CXCL1*, *CLEC11A* and *MKI67* while Schwann sub-clusters were identified as Schwann cell precursors (*SOX2* and *CDH19*), immature Schwann cells (*GAP43*) and myelinating Schwann cells (*MPZ*, *PLP1*).

Spatial transcriptomics and raw sequence data processing

Spatial transcriptomics was carried out using the 10x Genomics Visium platform. 10µm tissue sections from OCT-embedded fresh frozen human pancreases at 12, 15, 18 and 20 PCW were mounted onto Visium Spatial slides and the sections were permeabilized for 30min to release mRNAs, which bind to the spatially barcoded-oligos present in the underlying spots and reverse transcribed, according to the manufacturer's protocol. Libraries prepared from the cDNAs were sequenced on the Illumina NextSeq 2000 platform at >50,000 reads per spot generating >400M reads per section. Spaceranger software (version 3.1.0, 10x Genomics) was used to align and obtain raw counts from each of the spots on the Visium spatial transcriptomics slides against the GRCh38 human genome reference data.

10x Visium spatial transcriptomics data analysis

The spatial transcriptomics raw gene expression matrix, together with spatial location of spots and tissue H&E images, were used to create a Seurat object with a `Load10X_spatial` function. After normalization by `SCTransform`, we performed principal component analysis and reduced the dimensions to the top 20 principal components. Marker gene detection and differential gene expression were carried out using the `FindAllMarkers` function in Seurat. Genes that varied with locations *in situ* were identified using the `FindSpatiallyVariableFeatures` function, using default settings.

Spatial transcriptomics deconvolution and visualization

To spatially map the different developing human pancreas cells *in situ*, the combined scRNA-seq datasets were integrated with 10x Visium spatial transcriptomics data using the anchor-based integration pipeline in Seurat, which allowed the transfer of cell-type annotations from scRNA-seq to spatial transcriptomics. The cell type predictions from Seurat were loaded into a python package `stLearn`,³⁵ where the cell types in every spatial spot were annotated and visualized.

Spatial co-localization of receptor-ligand pairs

To investigate cell-cell interactions between endocrine progenitors and Schwann cells, we first annotated cell type diversity within each spot using cell type predictions in Seurat. We then identified significant ligand-receptor pairs between neighboring spots using `CellPhoneDB`,⁹² as implemented in `stLearn`. Ligand-receptor hotspots were defined as spatial regions containing high numbers of interacting cells and high ligand-receptor co-expression.

Spatial transcriptomics analysis using the Giotto package

To create a spatial network and identify spatial co-expression patterns, we reanalyzed the raw spatial gene expression matrix with Giotto,³⁰ and retained only spots that overlapped with the tissue area. We removed genes of low expression and low quality spots using `filterGiotto` with default parameters. Following normalization, we identified highly variable genes which were used to perform PCA analysis and shared-nearest neighbor identities were computed using the first 10 principal components. We performed Leiden clustering at a resolution of 0.4 to identify clusters of spots. We identified spatial co-expression modules and created spatial networks using the functions `detectSpatialCorGenes` and `createSpatialNetwork`, respectively. To identify distinct cell-cell interacting pairs, Giotto estimates the enrichment of the frequency that each interacting cell pairs are close to each other using the wrapper `CellProximityEnrichment`.³⁰ Spatial proximity enrichment or depletion was calculated by dividing the observed cell-cell proximity interactions by expected frequency of interactions, which is the average frequency obtained from 2000 spatial network simulations.

scRNA-seq cell-cell communication analysis

To infer cell-cell communication and identify ligands and receptors involved in endocrine progenitor/ Schwann cell interactions, we used the `Connectome v1.0.1` package.⁶¹ Using the `CellCellScatter` function, we identified the top signaling ligand-receptor pairs between endocrine progenitors and Schwann cells.

RNAscope multiplex single molecule in-situ hybridization and immunofluorescence

For RNAscope, fresh frozen human fetal pancreases were cryo-embedded in OCT and 10µm sections were fixed in 4% PFA for 15 min. Sections were dehydrated in ethanol before being treated with hydrogen peroxide and protease IV. Probes for *GAP43*, *NEUROG3*, *L1CAM* and *EPHB2* were hybridized for 2 hours at 40°C according to the manufacturer's protocols (Advanced Cell Diagnostics, document number 323100-USM) and signals were developed with Opal dyes (1:1500). Imaging was carried out with a 60x/1.4 NA oil objective using a Nikon A1 inverted confocal microscope. For immunofluorescence, formalin-fixed paraffin-embedded human fetal pancreases were de-waxed, sectioned at 5µm and immunoprobed for cytokeratin 19, insulin, glucagon,

ghrelin and carboxypeptidase A1 following standard protocols.⁹³ Imaging was carried out with a 20x or a 60x/1.4 NA oil objective using an A1 inverted confocal microscope.

Slingshot pseudotime trajectory analysis

Slingshot trajectory inference⁴⁰ was carried out on the endocrine and endocrine progenitor sub-clusters. To overcome graphical distortion seen in 2D UMAPs⁴¹ and unconnected topology in the endocrine cluster, we ran slingshot trajectory inference in 3D UMAP. The generalized additive model (GAM) was fit along the trajectories using tradeSeq algorithm⁸⁰ to identify differentially expressed genes.

Monocle 3 pseudotime trajectory analysis

Trajectory inference with Monocle 3⁴⁵ was carried out on the endocrine Seurat subset. We set NEUROG3-positive cells as the root of the trajectory and identified differentially expressed genes across the trajectories using the Moran's I test as implemented in Monocle 3.

Gene ontology enrichment analysis

Gene lists generated from the above-described analyses were used as inputs for gene ontology enrichment analysis, which was carried out on a web-based platform, Enrichr.⁶⁴

Spatial trajectory analysis

To investigate cellular trajectories in situ, we reprocessed the raw spatial data following documented protocols in stLearn.³⁵ Following Louvain clustering, we performed global and local pseudo-space-time trajectory analysis in stLearn which incorporates PAGA⁶² and a diffusion pseudotime method⁶³ to reconstruct trajectories using changes in transcriptional states between clusters of interest. Genes that were differentially upregulated or downregulated along the trajectories were determined by Spearman's rank correlation with a threshold of 0.3.